C. Maximum splitting

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: dp, greedy, math, number theory | https://codeforces.com/problemset/problem/870/C

You are given several queries. In the *i*-th query you are given a single positive integer n_i. You are to represent n_i as a sum of maximum possible number of composite summands and print this maximum number, or print -1, if there are no such splittings.

An integer greater than 1 is composite, if it is not prime, i.e. if it has positive divisors not equal to 1 and the integer itself.

Input

The first line contains single integer q ($1 \le q \le 10^5$) — the number of queries.

q lines follow. The (i + 1)-th line contains single integer n_i ($1 \le n_i \le 10^9$) — the i-th query.

Output

For each query print the maximum possible number of summands in a valid splitting to composite summands, or -1, if there are no such splittings.

Examples

Input
1 12

Output
3

Input
2 6 8

Output
1 2

Input
3 1 2 3

Output
-1 -1 -1

Note

12 = 4 + 4 + 4 = 4 + 8 = 6 + 6 = 12, but the first splitting has the maximum possible number of summands.

8 = 4 + 4, 6 can't be split into several composite summands.

1, 2, 3 are less than any composite number, so they do not have valid splittings.